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Regional growth, convergence, and spatial spillovers in India:
A reproducible view from outer space

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Abstract. This article demonstrates how a complete spatial econometric study can be published so that any reader can re-execute it in a browser. The demonstration extends Chanda and Kabiraj (2020, World Development), who used satellite nighttime light data as a proxy for economic activity across 520 districts in India. Every estimate, figure, and table reported below is generated by a Python notebook that runs in the cloud without any local installation. We extend their main findings on three fronts. First, we illustrate regional convergence patterns using an interactive tool for satellite imagery visualization. Second, we assess the degree of spatial dependence in their main econometric specification. Third, we employ a spatial Durbin model to measure the role of spatial spillovers in the convergence process. Our results indicate that spatial spillovers increase the estimated speed of regional convergence. Accounting for them raises the implied speed from about 3.0% to about 5.2% per year, which shortens the half-life of regional disparities from roughly 23 to 13 years. These conclusions are robust across seven definitions of the spatial weight matrix. We close by extending the same workflow beyond economic output, with an exploratory analysis of luminosity and cultural participation across Indian states, and by setting out the gaps in data, geography, and theory that this line of work must close next.

1 Introduction

Regional economic growth and convergence are central concerns in developing countries. They matter most in large federal states such as India, where spatial inequalities can threaten social cohesion and political stability. Studying convergence in such settings has long been difficult, because consistent economic data are rarely available below the state level. Satellite nighttime light data, used as a proxy for economic activity, have relaxed that constraint and enabled a growing literature on regional growth at granular geographic scales.

Chanda, Kabiraj (2020) used nighttime light data to document regional convergence across 520 districts in India between 1996 and 2010. Their analysis showed that poorer districts grew faster than richer ones, which suggests a reduction in spatial inequalities. Their econometric approach, however, did not model spatial dependence. The initial luminosity of neighboring districts enters their analysis only as a robustness control, with no spatially lagged dependent variable and no decomposition of direct and indirect impacts. The growth trajectory of a district may nevertheless depend not only on its own initial conditions but also on those of its neighbors.

This article extends the study of Chanda, Kabiraj (2020) in three methodological directions. First, we develop an interactive visualization tool for exploring the spatial and temporal patterns of regional convergence in nighttime lights. The tool helps identify converging regions and growth hotspots that static visualizations obscure. Second, we formally test for spatial dependence in both the dependent and the independent variables of the convergence equations. These tests show that spatial autocorrelation is a relevant feature of satellite data and of the regional convergence process. Third, we employ a spatial Durbin model to account for spatial spillovers and to quantify how neighbors influence the speed of regional convergence.

Our results contribute to the understanding of regional convergence in India on three fronts. First, interactive visualization reveals clear spatial patterns in both the initial distribution and the subsequent growth of nighttime lights. Second, formal tests of spatial dependence indicate that district-level economic trajectories are not independent of their neighbors. Third, the spatial Durbin model shows that catch-up across Indian districts is a neighborhood process and not only a district one. The indirect effect of initial luminosity is negative and statistically significant at the 10% level, so districts whose neighbors started from a lower base grew faster than their own initial conditions alone would predict.

These contributions have three dimensions, and it is worth stating which of them carries the most weight. The article is in part a replication and extension. It adopts the data, the sample, and the conditioning set of Chanda, Kabiraj (2020) unchanged, so that any difference in the estimates reflects the spatial specification and not a different research design. It is in part a substantive contribution to what is known about convergence across Indian districts, which the next paragraph sets out. Its primary contribution, however, is methodological. The article demonstrates that a spatial econometric study of regional convergence, running from raw satellite imagery through the construction of the weight matrix, the decomposition of direct and indirect impacts, and the robustness checks, can be published in a form that any reader can re-execute in a browser without installing anything. This journal has made the case for that form of publication in a special issue devoted to computational notebooks (Rowe et al. 2020), which includes a reproducible notebook for acquiring and processing satellite imagery (Chen et al. 2020). A Python notebook for processing nighttime lights data has also appeared here (Patnaik, Mendez 2024). More recently, Rey (2025) applied the same combination of Python and Quarto that we use here to the measurement of spatial inequality, and distributed the Quarto source of that article alongside its PDF and HTML versions. What this article adds to that line of work is a complete applied study rather than a tool or a demonstration dataset. Every estimate, figure, and table in the sections that follow is produced by one of the notebooks listed in Table 1, each of which runs in a browser.

That this pattern raises rather than lowers the measured speed of convergence is a substantive result and not merely a methodological one. That convergence estimates respond to spatial dependence is well established, but the direction of the response is not, and on that point the evidence divides. Allowing for spatial dependence lowers the estimated speed across European regions and Indonesian districts, and it reduces the estimated role of initial income across Indian states at a more aggregated level (Isla-Castillo et al. 2024, Santos-Marquez et al. 2022, Lolayekar, Mukhopadhyay 2019). Across Indian districts it raises the estimated speed, from about 3.0% to about 5.2% per year, because here the spillover works in the same direction as within-district catch-up rather than against it. What the article adds on this point is therefore a signed estimate of that effect for a specific setting, benchmarked against the comparable literature in the results section, rather than a further demonstration that spatial dependence matters.

Beyond these three contributions, we illustrate the wider potential of luminosity data with an exploratory analysis of how it relates to cultural participation across Indian states. That exercise relies on state-level data for an adjacent period, and on rank correlations rather than on the district-level convergence framework. We therefore present it as a demonstration of what the workflow makes possible rather than as additional evidence on the convergence process itself.

These findings also carry implications for methodology and policy. Methodologically, they suggest that in settings of this kind conventional non-spatial approaches understate the speed of regional convergence by failing to account for inter-district spillovers. From a policy perspective the difference is large enough to matter: the implied half-life of regional disparities falls from roughly 23 years to 13, which separates catch-up over a generation from catch-up within the horizon over which development programs are designed, funded, and evaluated. If the indirect effects reflect genuine spillovers, the benefits of place-based policies would extend beyond the districts they target and would raise their cost-effectiveness relative to what conventional estimates imply. The Indian evidence, however, also carries a warning: a district-targeted tax program has been found to raise activity partly by displacing it from neighboring districts that narrowly missed qualifying (Hasan et al. 2021), and displaced activity would appear in our framework as a spillover. We weigh both readings in the discussion.

This reproducible approach rests on Jupyter notebooks and the Quarto publishing system.[1] Together, these tools make every analytical step transparent. They also allow any reader to re-execute the full analysis from the raw data.

Every figure and table in this article is produced by one of the six computational notebooks listed in Table 1. The five analytical notebooks are written in Python and run directly in Google Colaboratory, which removes the usual barrier to reproduction: a reader needs no local installation, no environment configuration, and no software licenses. Opening a notebook through its Colab badge and clicking *Run all* downloads the raw data from the project repository and re-executes the complete analysis, regenerating every figure and table that the notebook contributes to this article. The remaining notebook, N1, hosts the interactive visualization together with its Google Earth Engine source code, which runs in the Earth Engine Code Editor. The notebooks, the data, and the manuscript source are available in the project repository at https://github.com/quarcs-lab/project2025s-py, and their rendered versions are embedded in the online edition of this article at https://quarcs-lab.github.io/project2025s-py/. We encourage readers to open them alongside the sections that follow.

Table 1: Computational notebooks underlying this article. Each analytical notebook carries a Colab badge that opens it in the browser, ready to run without any local setup.

Notebook	Content	Runs in
N1. View from outer space	Interactive visualization of regional luminosity and its Earth Engine source code	Earth Engine Code Editor
N2. Regional convergence	Bivariate convergence of luminosity growth on initial luminosity, with the implied speed and half-life	Google Colaboratory
N3. Spatial dependence	Spatial weight matrices, Global Moran's I, and LISA cluster maps	Google Colaboratory
N4. Spillover modeling	OLS and spatial Durbin estimates, the decomposition into direct and indirect impacts, and convergence speeds	Google Colaboratory
N5. Robustness	Model 4 re-estimated under seven spatial weight matrices	Google Colaboratory
N6. Spatial culture	Luminosity and cultural participation across Indian states	Google Colaboratory

[1] Jupyter notebooks integrate executable code, narrative text, and computational outputs within a single document. Quarto is an open-source publishing system that generates HTML, PDF, and Word versions from a single source file, which ensures consistency across all versions of a document.

Setting out what this approach can do also makes clear what it cannot do yet, and three constraints stand out. The first concerns the data: the DMSP-OLS series on which this literature rests is coarse, saturates in bright urban cores, and blurs light across administrative boundaries. The second concerns geography: a single satellite observes the entire planet, but it records only development that is illuminated, so the regions where lights are least informative are also those where conventional statistics are weakest. The third concerns theory: luminosity is now applied well beyond convergence analysis, yet it discriminates differences across places far better than changes over time, which limits what studies over short horizons can credibly claim. We return to each of these in the discussion, where they define a research agenda rather than a list of caveats.

The rest of this article is organized as follows. Section 2 describes the data and methods, including the use of nighttime light data as a proxy for economic activity and the extensions related to reproducible open science, interactive visualization, spatial dependence testing, and spillover modeling. Section 3 presents our empirical results, beginning with an interactive exploration of convergence patterns, followed by formal tests of spatial dependence and estimates of direct and indirect convergence effects from the spatial Durbin model. Section 4 draws out what the magnitude of the estimates implies for regional policy, sets out the limitations of relying on a single growth period, discusses an exploratory extension of the workflow beyond economic output, identifies the gaps in data, geography, and theory that remain open, and considers what reproducible practice contributes to this kind of research. Finally, Section 5 offers some concluding remarks.

2 Data and methods

2.1 Data: Nighttime lights as a proxy for economic activity

One of the key challenges in studying economic growth in developing countries such as India is the limited availability and reliability of data on aggregate economic activity below the state level. To address this problem, a growing body of literature pioneered by Henderson et al. (2012) has used satellite nighttime light data as a proxy for economic activity at subnational levels. This approach exploits the empirical correlation between the intensity of artificial light observed from satellites and the level of economic output on the ground. Chen, Nordhaus (2011) show that the proxy carries most information where official statistics are of low quality.

Nighttime light (hereafter NTL) data have been widely used to investigate economic growth and convergence across national and subnational regions. Applications include the effect of decentralization on convergence (Adhikari, Dhital 2021), the adjudication between national accounts and household surveys, which national accounts win (Pinkovskiy, Sala-i Martin 2016), and the global estimation of GDP per capita and spatial inequality (Lessmann, Seidel 2017). The Indian literature is equally varied, and it covers welfare programs, colonial institutions, demonetization, and the pandemic.[2] In the context of convergence studies, Chakravarty, Dehejia (2017) document significant regional disparities in India using NTL data and caution that the goods and services tax may further exacerbate them.

Our study builds on Chanda, Kabiraj (2020). Following their approach, we use the per capita growth in nighttime lights as the dependent variable and the initial nighttime lights per capita as the primary explanatory variable. For 520 districts in India, these variables are derived from NTL data released by the National Geophysical Data Center (NGDC), based on observations from the DMSP-OLS satellites over the period from 1996 to 2010. To mitigate the issue of top-coding, the NGDC released radiance-calibrated nighttime lights for eight specific years, using high magnification settings for low-light regions and low magnification settings for brightly lit areas. We use the radiance-calibrated series throughout.[3]

[2]Nighttime lights have been used in India to analyze the impact of public welfare programs (Cook, Shah 2022), the effects of colonial institutions (Jha, Talathi 2024), the impact of demonetization (Chanda, Cook 2022), and the effects of COVID-19 (Beyer et al. 2023).

[3]Following Chanda, Kabiraj (2020), our sample consists of 520 districts (out of a possible 593). There

2.2 Jupyter and Quarto for reproducible open science

Jupyter notebooks integrate executable code, explanatory narrative, and computational outputs within a single self-contained document (Kluyver et al. 2016), which is the practical expression of literate programming (Knuth 1984). We employ notebooks running a single Python kernel to document our data processing, analysis, and visualization steps, so that each result can be traced back to the code that produced it. Relying on one open-source toolchain also means that the entire pipeline can be re-executed without proprietary software: every analytical notebook opens in Google Colaboratory and runs end to end in the browser, installing its own dependencies and fetching the raw data from the project repository.

Quarto extends these notebooks into a manuscript framework (Allaire et al. 2024). The figures and tables in this article are programmatically embedded from specific notebook cells, so that every empirical claim in the text is traceable to the computation behind it. A single source file generates the HTML, PDF, Word, and JATS XML versions of the article without discrepancies between them. The raw data and code are available in a public repository, so that other researchers can verify the analysis and build upon it (Peng 2011).

2.3 Google Earth Engine for interactive spatial visualizations

Satellite data have opened new dimensions of economic measurement (Donaldson, Storeygard 2016). Interactive visualizations of nighttime lights imagery offer further methodological advantages over static representations, especially when the analysis concerns temporal and spatial heterogeneity in economic activity. Google Earth Engine (GEE) lowers the computational and technical barriers to creating such visualizations and provides an accessible platform for analyzing satellite data (Gorelick et al. 2017). Its browser-based development environment supports interactive web applications without specialized local software, and its extensive catalog of pre-processed datasets, including the DMSP-OLS and VIIRS nighttime lights collections, reduces the storage and processing burden (Taminia et al. 2020). For the purposes of this article, the platform allows readers to track the evolution of light intensity across regions over time, and therefore to observe catch-up growth directly, as initially dim regions converge toward the luminosity levels of their brighter counterparts.

2.4 Regional convergence modeling

In neoclassical growth models, the per capita growth rate is predicted to be negatively correlated with the initial endowment of a region, primarily because of diminishing returns to capital accumulation (Solow 1956). Poorer regions, assuming similar technology and preferences, are expected to experience higher growth rates than their wealthier counterparts. Over the long run, regions with similar characteristics should therefore converge to a common steady state. We examine this convergence hypothesis within the growth regression framework of Barro, Sala-i Martin (1992), and Equation 1 presents the convergence process in its simplest unconditional form.

$$g_t = \beta_1 x_{t-1} + \varepsilon_t \tag{1}$$

where g_t represents an N-by-1 vector of observations on per-capita NTL growth for each of the N regions over the period t. The vector x_{t-1} represents an N-by-1 vector of observations on the initial (log) level of per-capita NTL. The parameter β_1 is a regression coefficient that indicates the direction and strength of regional convergence. A negative value of β_1 would suggest that regions with lower initial NTL levels grow faster, consistent with the convergence hypothesis. Finally, ε_t represents a vector of idiosyncratic error terms.

were 593 districts in 2001, which rose to 640 districts in the 2011 census. To match districts across the two census files, we merged newly split districts back with their parent districts in the 2011 census. Eight of the 47 new districts were created by splitting areas from multiple parent districts. Those new districts along with their multiple-origin districts were dropped from our sample. We dropped all the districts in the state of Assam where more than 50% of districts were created in that manner.

To facilitate interpretation, the estimated convergence coefficient can be translated into an annual speed of convergence and a half-life (Barro, Sala-i Martin 1992). The dependent variable is the average annual growth rate over a period of T years, so the implied speed of convergence is $\lambda = -\ln(1 + \beta_1 T)/T$. The corresponding half-life, which is the time required to close half of the initial gap to the steady state, is $\ln(2)/\lambda$. In our application $T = 14$ years, and the same transformation is applied to the total effect of initial luminosity in the spatial models.

Because this simple framework contains no control variables, it implies that districts converge to a common steady state. Regions may nevertheless differ in geography, socio-economic conditions, and policy implementation. To account for these differences, we include state fixed effects that control for state-specific institutions and policies, and we add a range of geo-climatic controls alongside district-specific conditions related to demographics, human capital, and infrastructure. Equation 2 summarizes this conditional convergence framework.

$$g_t = \beta_1 x_{t-1} + X_t \alpha + \varepsilon_t \tag{2}$$

Here, the matrix X_t is an N-by-k collection of observations on control variables (including state fixed effects) for each district in our sample. The vector α, with dimensions $k \times 1$, captures the regression coefficients for these variables. The conditioning set is adopted unchanged from Chanda, Kabiraj (2020), and this is deliberate: because the present analysis extends that study rather than replacing it, holding the controls fixed ensures that any difference between our estimates and theirs is attributable to the spatial specification and not to a different choice of covariates. The 16 district-level controls fall into three groups.[4] Geography and agro-climate capture slow-moving determinants of the steady state that do not vary over the sample period. Demographics and human capital capture the factor endowments through which districts accumulate. Infrastructure captures access to the grid and to markets, which matters here because luminosity is itself partly a measure of electrification, so the conditional models compare districts that began the period with similar levels of grid access. The demographic, human-capital, and electrification controls are measured in 1996, at the start of the growth period, and are therefore predetermined with respect to subsequent growth. The paved-roads variable is measured in 2000. The full list of control variables, with their measurement years and summary statistics, is given in Table A.1 of Chanda, Kabiraj (2020).

Both specifications are estimated on a single long difference: average annual growth between 1996 and 2010, regressed on conditions at the start of that period. This is the design of Chanda, Kabiraj (2020) and we retain it, but it should be stated plainly what it does and does not allow. It identifies the average rate at which districts converged over fourteen years, but it cannot speak to how that rate evolved within the period. We return to this limitation, and to what a panel design would require, in the discussion.

2.5 Spatial dependence testing

The analysis of regional convergence using nighttime light data requires explicit consideration of spatial dependencies across districts. Spatial dependence is the tendency for observations at nearby locations to be more similar than those at distant locations, and it can arise from spatial spillovers, shared geographic conditions, or inter-regional economic linkages (Anselin 1988). If present and unaccounted for, spatial dependence can lead to biased parameter estimates and invalid inference in standard regression models. To formally test for such spatial relationships, we employ the Global Moran's I statistic and Local Indicators of Spatial Association (LISA), applied to the main variables of Equation 2.

The Global Moran's I statistic, originally proposed by Moran (1950), is the most widely used measure of spatial autocorrelation. It quantifies the overall degree of spatial

[4] Geography and agro-climate comprise agricultural suitability, rainfall, temperature, a malaria ecology index, terrain ruggedness, latitude, and distance to the coast. Demographics and human capital comprise the rural population share, log population density, the scheduled caste and scheduled tribe shares, the working population share, and the literate and higher-educated shares. Infrastructure comprises the share of households with an electricity connection and the log number of households with access to paved roads.

clustering among geographic units. It can be expressed as Equation 3:

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij} z_i z_j}{\sum_i z_i^2} \tag{3}$$

where z_i represents the deviation of observation i from the mean, w_{ij} denotes the spatial connection (weight) between units i and j, and n is the total number of observations. The statistic typically ranges from -1 to $+1$, with positive values indicating that similar values tend to be located near each other.[5] Statistical significance is assessed through a permutation-based inference approach.[6]

While the Global Moran's I provides a single summary measure of overall spatial dependence, it does not reveal where significant clusters or outliers are located. To address this limitation, Anselin (1995) proposed Local Indicators of Spatial Association, which decompose the global statistic into contributions from each individual observation. The local Moran's I for observation i is defined in Equation 4:

$$I_i = z_i \sum_j w_{ij} z_j \tag{4}$$

where the summation is over the neighbors of i as defined by the spatial weight matrix. This local statistic classifies each observation into one of four categories, according to the relationship between the value at a location and the values of its neighbors.[7] The High-High (HH) and Low-Low (LL) categories identify spatial clusters, while the High-Low (HL) and Low-High (LH) categories identify spatial outliers. Statistical significance of each local statistic is assessed through conditional permutation tests, and only statistically significant locations (typically at $p < 0.05$) are reported in the LISA cluster maps.

The specification of the spatial weights matrix $\mathbf{W}$ is important for capturing the underlying spatial structure. We employ a six-nearest-neighbors (6NN) weight matrix, so that for each district the six nearest districts by centroid distance are identified as its neighbors. This specification ensures a fixed number of connections per district regardless of the heterogeneity in district sizes across the country:

$$\mathbf{W} = \begin{bmatrix} w_{11} & w_{12} & w_{13} & \cdots & w_{1n} \\ w_{21} & w_{22} & w_{23} & \cdots & w_{2n} \\ \vdots & \vdots & \vdots & w_{ij} & \vdots \\ w_{n1} & w_{n2} & w_{n3} & \cdots & w_{nn} \end{bmatrix}$$

where $w_{ij} = 1$ if district j is among the six nearest neighbors of district i, and $w_{ij} = 0$ otherwise. Following standard practice, we row-normalize the weights matrix so that each row sums to one.[8] The choice of $k = 6$ provides a balance between capturing local spatial interactions and avoiding the inclusion of geographically distant districts that are unlikely to exert meaningful economic influence. This choice is also supported empirically: among the seven weight matrices considered in Table 4, the 6NN specification attains the lowest AIC, indicating the best fit to the spatial structure of the data.

The detection of significant spatial dependence would justify the use of spatial econometric techniques. In particular, Ertur, Koch (2007) argues that the spatial Durbin model (SDM) can appropriately account for spatial dependence in the convergence process, and Fischer (2011) derives the same specification as the reduced form of a spatially interdependent growth model. This methodological choice allows us to distinguish between direct effects of district characteristics and indirect effects operating through spatial channels (LeSage, Pace 2009).

[5] Values near zero suggest spatial randomness, and negative values indicate spatial dispersion, where dissimilar values are neighbors.

[6] The observed statistic is compared against a reference distribution generated by randomly reassigning values across locations.

[7] High-High (HH) indicates a high-value location surrounded by high-value neighbors, and Low-Low (LL) indicates a low-value location surrounded by low-value neighbors. High-Low (HL) is a spatial outlier where a high-value location is surrounded by low-value neighbors, and Low-High (LH) is the opposite spatial outlier.

[8] Row-normalization makes the spatial lag $\mathbf{Wx}$ the average value among the neighbors of a district.

2.6 Spatial spillover modeling

Our spatial spillover modeling builds upon the spatially augmented Solow model of Ertur, Koch (2007), developed for countries, and its regional Mankiw-Romer-Weil extension by Fischer (2011). Both extend the traditional Solow framework to account for technological interdependence across economies. We apply that framework to an economy of N subnational regions, each characterized by the Cobb-Douglas production function with constant returns to scale in Equation 5:

$$Y_{it} = A_{it} K_{it}^{\alpha_K} H_{it}^{\alpha_H} L_{it}^{1-\alpha_K-\alpha_H} \tag{5}$$

where Y_{it} represents output, K_{it} physical capital, H_{it} human capital, L_{it} labor force, and A_{it} the level of technological knowledge for region i at time t. The parameters α_K and α_H denote the output elasticities with respect to physical and human capital, respectively. A key innovation of this framework is the modeling of technological knowledge in Equation 6, which incorporates both internal and external factors:

$$A_{it} = \Omega_t k_{it}^{\theta} h_{it}^{\phi} \prod_{j \neq i}^{N} A_{jt}^{\rho W_{ij}} \tag{6}$$

This specification captures three distinct components of technological progress:

- An exogenous component (Ω_t) representing the common stock of knowledge across regions
- An embodied component ($k_{it}^{\theta} h_{it}^{\phi}$) reflecting technology embedded in physical and human capital per worker
- A spatial component ($\prod_{j \neq i}^{N} A_{jt}^{\rho W_{ij}}$) capturing technological interdependence between regions

Based on this theoretical framework, Ertur, Koch (2007) derived a spatial Durbin model that accounts for spillovers in the convergence process, with the human-capital terms following Fischer (2011). The model augments the conditional specification of Equation 2 with spatial lags of the dependent variable, of initial luminosity, and of the control variables. It can be compactly written in matrix notation as Equation 7:

$$g_t = \beta_1 x_{t-1} + X_t \alpha + \beta_2 W x_{t-1} + W X_t \gamma + \rho W g_t + \varepsilon_t \tag{7}$$

The vectors g_t, x_{t-1} and ε_t, the matrix X_t, and the parameters β_1 and α are defined as in Equation 2 above. The new terms are the spatial lags: $W x_{t-1}$ and $W g_t$ are N-by-1 vectors of neighborhood averages of initial luminosity and of growth, while $W X_t$ is the corresponding $N \times k$ matrix for the control variables. The coefficients β_2, γ and ρ measure how strongly each of these neighborhood averages enters the growth of a district.[9]

3 Results

3.1 Regional convergence: An interactive exploration from outer space

Before presenting the regression results, we visually illustrate the concepts of growth and convergence in nighttime lights. First, we create an interactive map of India displaying regional luminosity in 1996 and 2010.[10] Second, we examine absolute convergence by constructing a convergence scatterplot, which relates per-capita growth rates in nighttime lights over 1996–2010 to initial per-capita nighttime light levels in 1996 for each district. Finally, we present case studies of some of the poorest regions in the country. These cases show an increase in nighttime lights that visually aligns with the convergence hypothesis.

Figure 1 presents static maps of luminosity in the initial and final years, 1996 and 2010, captured from our interactive application. The maps show a noticeable increase in brightness across most parts of India in 2010. Because nighttime lights serve as a proxy

[9]We denote the spatial autoregressive parameter by ρ to avoid any confusion with λ, which denotes the implied speed of convergence throughout this article.

[10]The interactive web application is available at https://bit.ly/india-rc-ntl. It was developed using Google Earth Engine, and the source code is available in the View from outer space notebook.

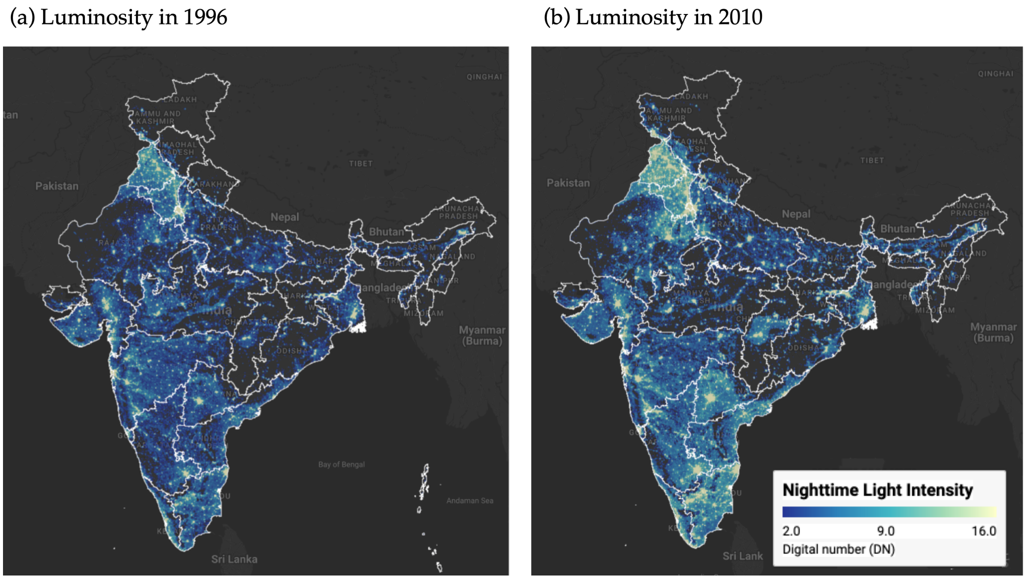


Figure 1: Regional luminosity in India: 1996 vs 2010 Notes: Luminosity is measured in radiance-calibrated digital number (DN) values from DMSP-OLS satellites. Interactive web application available at https://bit.ly/india-rc-ntl. Source: Authors' visualization using pre-processed luminosity images from the Earth Observation Group (NOAA/NCEI). See View from outer space notebook for source code.

for economic activity, this increase in luminosity reflects economic growth over the period. Figure 2 then illustrates the relationship between per-capita growth in nighttime lights and initial per-capita nighttime light. The scatterplot shows an inverse relationship: the estimated β-convergence coefficient of about -0.02 implies an annual speed of convergence of roughly 2.3% and a half-life of about 30 years, consistent with the seminal finding of Barro, Sala-i Martin (1992).

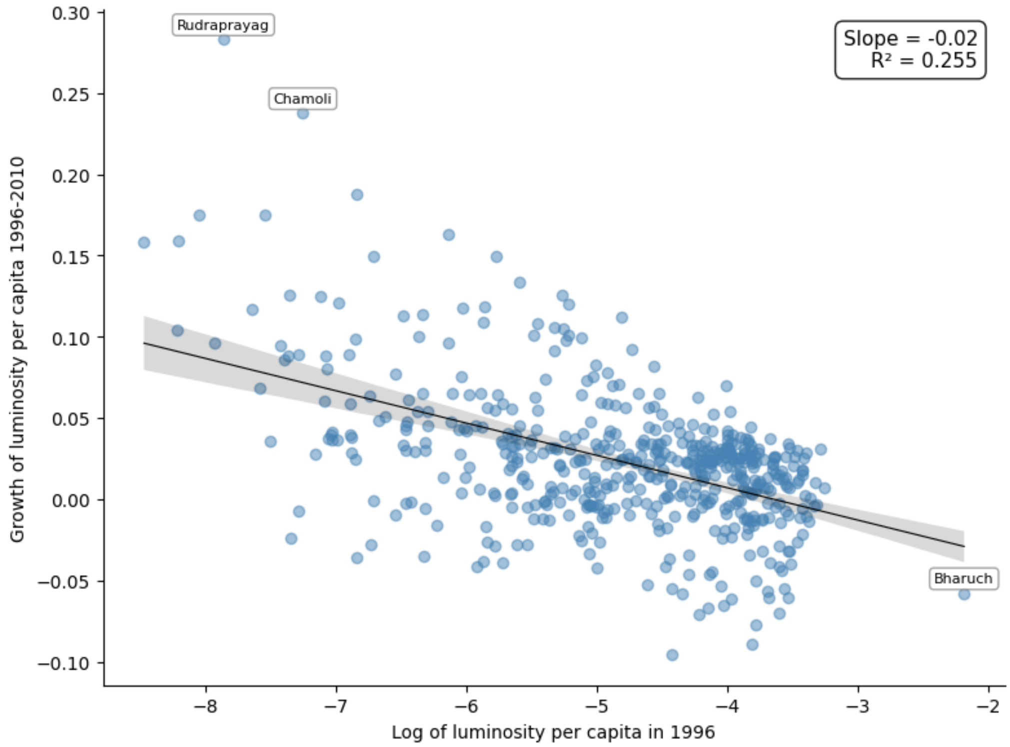


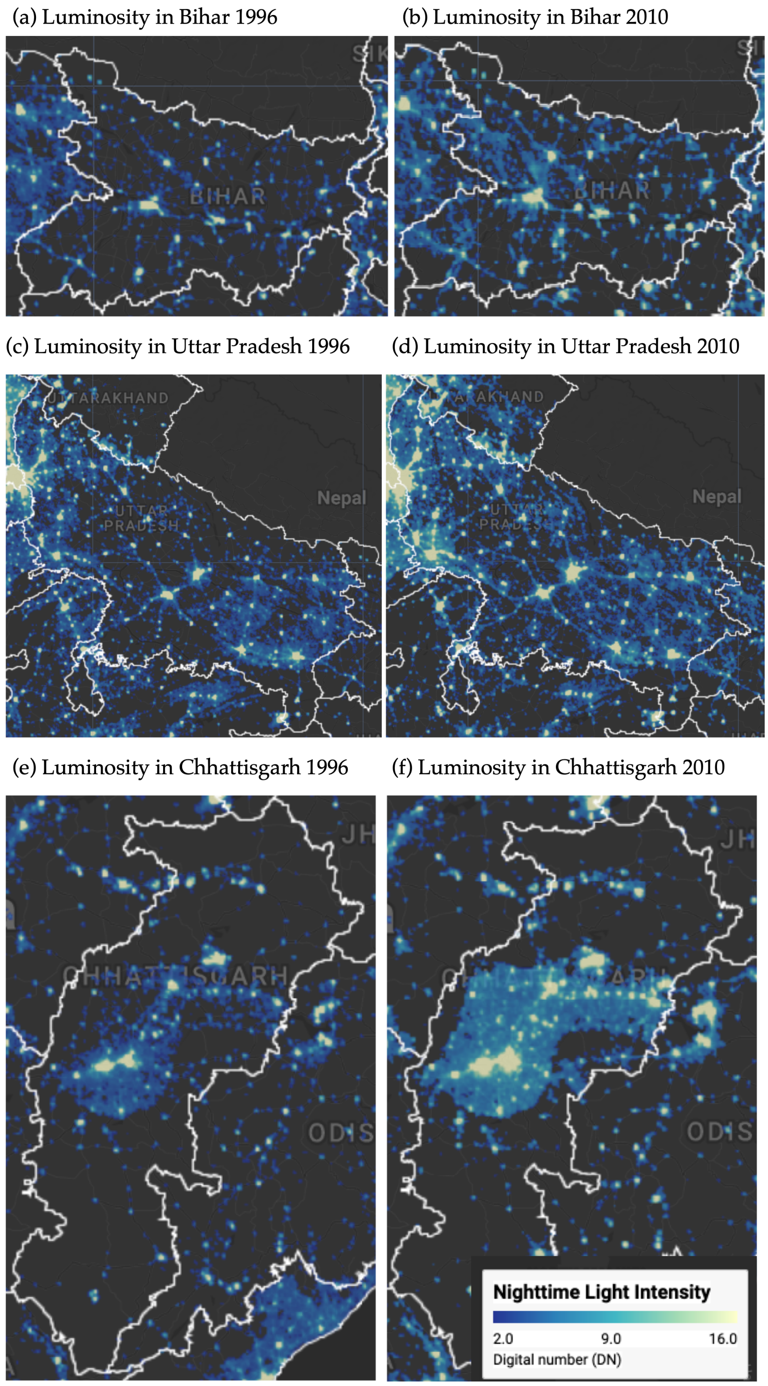
Figure 2: Regional luminosity convergence across districts in India Notes: Each point represents one of the 520 districts. The regression line shows the estimated beta-convergence relationship. Outlier districts are labeled. Source: Data from Chanda and Kabiraj (2020). See Regional convergence notebook for source code.

Next, we examine case studies of three economically disadvantaged states in India to illustrate their growth patterns over the study period. Among these, Bihar is the poorest, with a per-capita income at 31.2% of the national average, while Uttar Pradesh and Chhattisgarh stand at 55.3% and 59.9% respectively.[11] Figure 3 displays the change in luminosity in these three states during our study period. Although these states remain among the poorest, there is a noticeable increase in luminosity over the course of the study.

3.2 Spatial dependence is a feature of the convergence process

Before proceeding with the formal econometric analysis, we examine the spatial distribution of the variables under study. Choropleth maps are a natural tool for this purpose, because they show how a variable of interest varies across geographic units and reveal patterns that summary statistics alone would hide. Figure 4 presents the spatial distribution of initial luminosity in 1996 and of the subsequent growth rate of luminosity over the 1996–2010 period across the 520 districts in our sample.

[11] The figures are relative per capita income for 2000–01, the census year closest to the start of our study period, measured as per capita net state domestic product divided by per capita net national income. They are taken from Table 4 of Sanyal and Arora, "Relative Economic Performance of Indian States: 1960-61 to 2023-24", Economic Advisory Council to the Prime Minister, Working Paper EAC-PM/WP/31/2024, September 2024.



(a) Luminosity in Bihar 1996 (b) Luminosity in Bihar 2010

(c) Luminosity in Uttar Pradesh 1996 (d) Luminosity in Uttar Pradesh 2010

(e) Luminosity in Chhattisgarh 1996 (f) Luminosity in Chhattisgarh 2010

Figure 3: Some illustrative examples of regional convergence Notes: Luminosity is measured in radiance-calibrated digital number (DN) values from DMSP-OLS satellites. Interactive web application available at https://bit.ly/india-rc-ntl. Source: Authors' visualization using pre-processed luminosity images from the Earth Observation Group (NOAA/NCEI). See View from outer space notebook for source code.

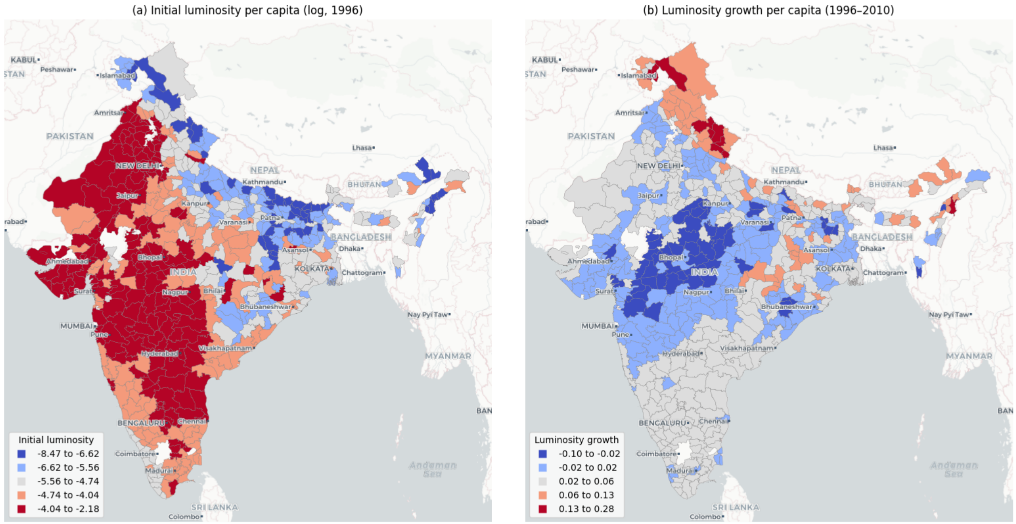


Figure 4: Spatial distribution of initial luminosity and luminosity growth Notes: Districts are classified into five categories using Fisher-Jenks natural breaks. Panel (a) shows log of luminosity per capita in 1996. Panel (b) shows luminosity growth per capita over 1996–2010. Source: Data from Chanda and Kabiraj (2020). See Spatial dependence notebook for source code.

The choropleth maps in Figure 4 reveal a distinct spatial pattern. Panel (a) shows that initial luminosity levels are concentrated in specific geographic corridors, with higher values clustered in western and north-western India, while large portions of eastern and northern India exhibit markedly lower luminosity. Panel (b), which displays the growth rate of luminosity per capita over the study period, presents a pattern that is largely the inverse of the initial distribution. Districts that were initially bright tend to exhibit lower growth rates, whereas districts that were initially dim tend to grow at a faster rate. This spatial inversion provides a first visual indication that regional convergence in India may have an important spatial dimension.

While the choropleth maps suggest the presence of spatial structure in both variables, a formal analysis of spatial dependence requires a definition of the neighborhood of each district. This neighborhood structure is specified through a spatial weight matrix, which encodes the connectivity between geographic units. In this study, we adopt a six nearest neighbors weight matrix, so that for each of the 520 districts the six nearest districts by centroid distance are identified as its neighbors. This connectivity structure is visualized as a network in Figure 5, where each node represents a district centroid and each edge connects a district to one of its six nearest neighbors.

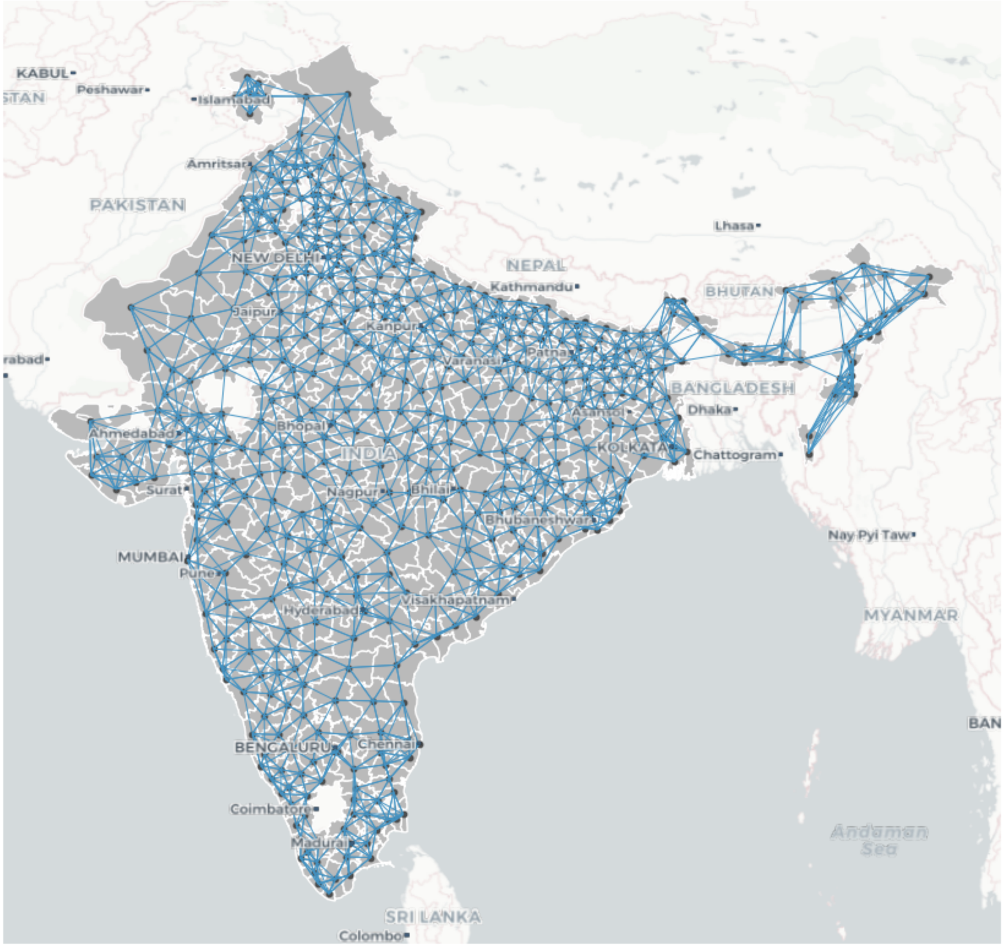


Figure 5: Spatial connectivity structure based on six nearest neighbors Notes: Each node represents a district centroid. Each edge connects a district to one of its six geographically closest neighbors. The weight matrix is row-standardized. Source: Data from Chanda and Kabiraj (2020). See Spatial dependence notebook for source code.

With the neighborhood of each district defined, we can formally assess the degree of spatial dependence in the variables. Spatial dependence can be understood through two complementary concepts: the overall degree of clustering in the data, and the specific locations of statistically significant clusters and spatial outliers. The Global Moran's I statistic captures the first of these concepts. In our data, the Moran's I for initial luminosity per capita is 0.73 ($p = 0.001$), which indicates strong positive spatial autocorrelation: districts with high (low) initial luminosity tend to be surrounded by districts that are also bright (dim). The Moran's I for luminosity growth is 0.60 ($p = 0.001$), which confirms that the growth rates of neighboring districts are also significantly correlated. Both the initial level and the subsequent growth of luminosity therefore exhibit substantial spatial clustering.

We now turn to the LISA statistics, which locate the significant clusters and outliers behind these global measures. Figure 6 presents the Moran scatterplot and LISA cluster map for initial luminosity per capita. The cluster map identifies distinct geographic concentrations. HH clusters mark the most luminous regions and their bright neighbors, while LL clusters highlight contiguous areas of low luminosity. These are predominantly in eastern and northern India, in Bihar, eastern Uttar Pradesh, and Jharkhand, together with the Himalayan north and the northeast.

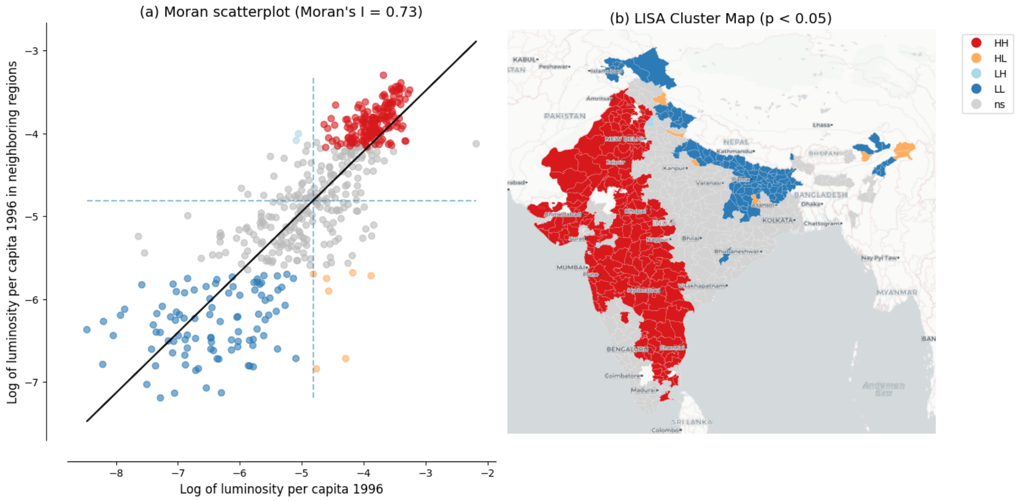


Figure 6: Spatial dependence in the initial level of luminosity Notes: Panel (a) shows the Moran scatterplot with Global Moran's I statistic. Panel (b) shows the LISA cluster map with statistically significant clusters at $p < 0.05$ based on 999 permutations. Source: Data from Chanda and Kabiraj (2020). See Spatial dependence notebook for source code.

The same LISA analysis applied to luminosity growth rates reveals a geographic pattern that is largely the inverse of the initial luminosity clusters, as shown in Figure 7. Districts that formed HH clusters in initial luminosity tend to appear as LL clusters in luminosity growth, and the dimmest clusters tend to emerge as HH clusters in growth.[12] This spatial inversion between the initial level and the subsequent growth of luminosity is the spatial signature of the convergence process: red regions in the initial luminosity map become blue regions in the growth map, and vice versa. These local spatial patterns provide visual evidence that spatial dependence is a prominent feature of the regional convergence process observed in India, and they motivate the use of spatial econometric methods.

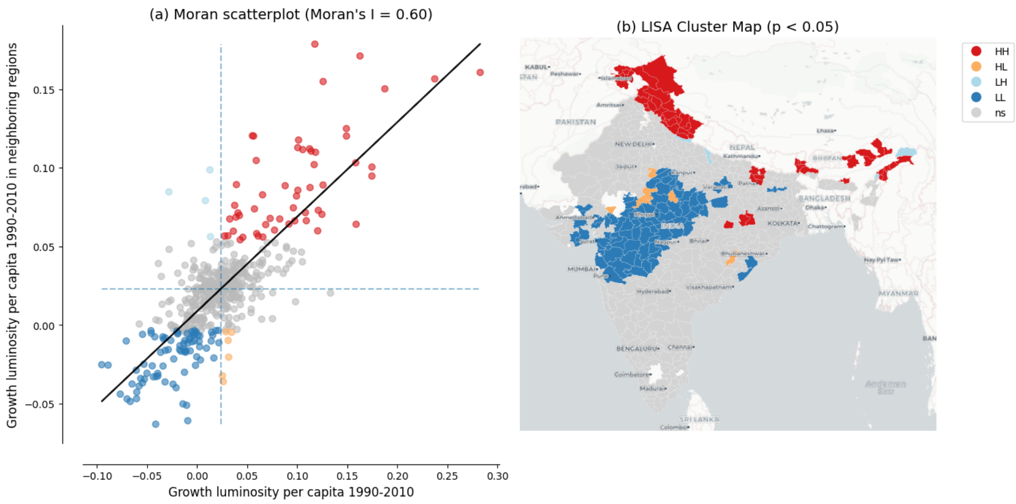


Figure 7: Spatial dependence in the growth rate of luminosity Notes: Panel (a) shows the Moran scatterplot with Global Moran's I statistic. Panel (b) shows the LISA cluster map with statistically significant clusters at $p < 0.05$ based on 999 permutations. Source: Data from Chanda and Kabiraj (2020). See Spatial dependence notebook for source code.

[12] The initially prosperous areas therefore experienced relatively slower growth over the study period, while the initially lagging areas grew faster.

3.3 Evidence of spatial spillovers in regional convergence

The regression results of Table 2 provide evidence of both unconditional and conditional convergence across districts in India. Both conventional ordinary least squares (OLS) and spatial econometric approaches indicate significant negative relationships between initial luminosity levels and subsequent growth rates. The direct effects, which represent within-district convergence, remain negative and of similar order across specifications, ranging from -0.020 to -0.026. This consistency across specifications and estimation methods suggests that poorer districts are catching up to their wealthier counterparts, even after controlling for district characteristics and state-level fixed effects.

Table 2: Unconditional and conditional convergence across districts.

	Model 1		Model 2		Model 3		Model 4	
	OLS	SDM	OLS	SDM	OLS	SDM	OLS	SDM
Direct	-0.020***	-0.026***	-0.022***	-0.021***	-0.025***	-0.026***	-0.025***	-0.025***
	(0.002)	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)
Indirect	–	0.004	–	-0.001	–	-0.015*	–	-0.013*
		(0.006)		(0.005)		(0.009)		(0.007)
Total	-0.020***	-0.022***	-0.022***	-0.022***	-0.025***	-0.041***	-0.025***	-0.037***
	(0.002)	(0.006)	(0.003)	(0.005)	(0.003)	(0.009)	(0.003)	(0.007)
Controls	No	No	No	No	Yes	Yes	Yes	Yes
State FE	No	No	Yes	Yes	No	No	Yes	Yes
AIC	-1945	-2292	-2409	-2468	-2211	-2358	-2465	-2501

The progression from unconditional to conditional specifications reveals how the estimated convergence process is shaped by the inclusion of additional covariates. In the OLS specifications, the direct effect is -0.020 in Model 1 and -0.022 in Model 2, and it strengthens to -0.025 once district characteristics are added in Models 3 and 4. Omitting these characteristics therefore attenuates the estimated speed of convergence. The spatial Durbin direct effects match this pattern in the conditional models, at -0.026 in Model 3 and -0.025 in Model 4.[13] Once we account for structural differences across districts, the underlying tendency for poorer districts to catch up becomes more pronounced. State fixed effects absorb unobserved state-level heterogeneity, such as differences in governance and institutional quality, that might otherwise confound the convergence relationship. They tighten the standard errors and moderate the spatial total effect rather than strengthening it further.

The spatial Durbin model also reveals spatial spillover effects that are not captured by traditional OLS estimations. These indirect effects, which capture the influence of the initial conditions of neighboring districts on the growth rate of a district, follow a notable pattern across specifications. In the unconditional models (Models 1 and 2), the estimated indirect effects are statistically insignificant and erratic, negative in Model 2 but positive in Model 1.[14] Once district-level controls are introduced (Models 3 and 4), the indirect effects become both larger in magnitude and statistically significant at the 10% level, reaching -0.015 in Model 3 and -0.013 in our most comprehensive Model 4. The spatial channels of convergence therefore become discernible only after accounting for district-specific characteristics.

[13] The impact decomposition is unreliable in Model 1, where the spatial autoregressive parameter is very large, at about 0.80, which inflates the direct effect to -0.026. In Model 2 the indirect effect is indistinguishable from zero.

[14] Without controls, the strong residual spatial dependence confounds the spillover estimates with omitted variables.

The total impact of initial conditions on growth, combining both direct and spillover effects, is substantially larger when we account for spatial dependence. In our fully specified model (Model 4), the total convergence effect in the spatial Durbin model (-0.037) is approximately 51% larger in magnitude than the OLS estimate (-0.025). Translated into an annual speed of convergence, this raises the implied speed from about 3.0% under OLS to about 5.2% under the SDM, and it shortens the implied half-life from roughly 23 to 13 years. The gap is even larger in Model 3 (SDM total -0.041 against OLS -0.025, or 67%), but the Model 3 and Model 4 spillover estimates are statistically indistinguishable, so we anchor our interpretation on the preferred Model 4. These differences indicate that in this setting conventional non-spatial approaches understate the speed of regional convergence by failing to capture the additional convergence channels created through spatial spillovers.

Table 3: Implied annual speed of convergence and half-life by model, from the OLS and SDM total effects of initial luminosity.

Model	OLS speed	SDM speed	OLS half-life	SDM half-life
Model 1	2.3%	2.6%	30 yr	27 yr
Model 2	2.6%	2.7%	27 yr	26 yr
Model 3	3.0%	6.2%	23 yr	11 yr
Model 4	3.0%	5.2%	23 yr	13 yr

Table 3 reports the implied speeds and half-lives for all four specifications. The spatial premium is concentrated in the conditional models. In Models 1 and 2 the OLS and SDM speeds are nearly identical, whereas in Models 3 and 4 the SDM raises the implied speed from 3.0% to 6.2% and 5.2%, shortening the half-life from 23 years to 11 and 13 years respectively.[15]

The model fit statistics further support the spatial econometric approach. The Akaike Information Criterion (AIC) consistently favors the spatial Durbin model over OLS across all four specifications, and the most comprehensive model (Model 4 SDM) achieves the lowest AIC value of -2501 compared to -2465 for the corresponding OLS.[16] Overall, these results suggest that regional convergence in India operates not only through district-specific factors but also through spatial interactions between neighboring districts. In this setting, ignoring these interactions leads to both underestimated convergence speeds and inferior model performance.

A word of caution is in order about how the indirect effects should be read. The spatial Durbin model identifies spatial dependence in reduced form: it establishes that the growth of a district covaries with the initial conditions of its neighbors, conditional on its own characteristics, but it does not establish why. Several distinct processes generate the same reduced-form pattern. Neighboring districts share geographic and agro-climatic conditions. They are often governed by the same state, and so inherit common institutional and policy environments that our state fixed effects absorb only in part. They are connected by infrastructure whose effects do not stop at administrative boundaries. Any omitted determinant of growth that is itself spatially clustered will also be absorbed into the spatial terms of the model. The measurement issues discussed below point in the same direction, since blooming spreads light across district boundaries and can raise the apparent correlation between neighbors for reasons unrelated to economic interaction. We therefore read the indirect effects as evidence that spatial structure is a feature of the convergence process, and not as an estimate of the causal effect of the conditions of one district on the growth of another. Establishing which channels are at work, and whether they are causal, would require research designs built for that purpose, a point we return to in the discussion.

[15] The unconditional speeds are 2.3% against 2.6% in Model 1 and 2.6% against 2.7% in Model 2.

[16] The improvement from incorporating spatial structure is most pronounced in the unconditional specification, where the AIC drops by 347 units when moving from OLS to SDM in Model 1, reflecting the large amount of spatial dependence left unmodeled by ordinary regressions. Even in the fully specified model, where state fixed effects already absorb much of the spatial heterogeneity, the SDM retains a meaningful advantage.

3.4 How these estimates compare with the convergence literature

The speeds reported above can be placed against three sets of benchmarks. The first is other work on Indian districts using the same proxy. Beyer, Yamada (2020) study district-level convergence in India with nighttime lights for 2013–2019 and report an absolute convergence rate of 2.4% and a conditional rate of 3.7%. Our own non-spatial estimates, 2.3% in the bivariate relationship and 3.0% in the conditional OLS specification, are close to these figures, even though the two studies cover different periods and rely on different generations of the sensor. That two independent applications of the same measurement strategy recover similar non-spatial convergence rates is reassuring about the proxy, and it locates our contribution in what the spatial specification adds rather than in the baseline itself.

The second comparison is with convergence estimated from luminosity elsewhere. Basher et al. (2023) find absolute convergence across the 64 districts of Bangladesh at 4.57% per year, implying a half-life of 15 years, which is close to the 5.2% and 13 years implied by our preferred spatial model. Carrington, Jiménez-Ayora (2021) report convergence across Ecuadorian provinces at a speed approximating the 2% benchmark. Kacou (2022), using night-light-based inequality proxies for 49 African countries, instead finds multiple equilibria in within-country regional inequality, with high-inequality countries showing no tendency to converge toward low-inequality ones. Read against the wider regional literature, our non-spatial estimates sit near the familiar 2% regularity while the spatial estimates sit above it.[17] That regularity should not be treated as a law: a meta-analysis of some 600 published estimates concludes that it is misleading to speak of a natural convergence rate, since estimates produced by different specifications are drawn from different populations (Abreu et al. 2005). These comparisons therefore locate our estimates rather than validate them.

The third comparison bears most directly on what this article adds. That convergence estimates are sensitive to spatial dependence is well established, but the direction in which allowing for it moves the estimated speed is not. Our result, an implied speed that rises from 3.0% to 5.2% once the spatial structure is modeled, agrees with Diop (2018), who finds that the estimated speed increases substantially in a spatial Durbin specification and concludes that earlier estimates may have been considerably understated. Other studies find the opposite. Isla-Castillo et al. (2024) report that allowing for spatial dependence lowers the estimated convergence speeds of European regions, and Santos-Marquez et al. (2022) conclude that neighbor effects have slowed the pace of convergence across Indonesian districts. Within India, Lolayekar, Mukhopadhyay (2019) find that the estimated impact of initial income across states is considerably smaller once spatial dependence is taken into account, and that states diverge rather than converge.

The arithmetic behind these differences is straightforward in our own case. The indirect effect carries the same negative sign as the direct effect, so the two reinforce one another and the total effect exceeds the OLS coefficient in magnitude. Where the initial conditions of neighbors enter with the opposite sign, or where the spatial terms absorb variation that the non-spatial model had attributed to initial luminosity, the same exercise reduces the estimated speed. The substantive point is therefore not that spatial dependence matters, which is already known, but that in this setting it works in the direction of faster measured convergence. The contrast with Indian states at a more aggregated level further suggests that the scale of analysis is part of what determines the answer.

3.5 Robustness to alternative spatial weight matrices

The spillover results above use a six-nearest-neighbor (6NN) spatial weight matrix. To assess their sensitivity to this choice, we re-estimate the preferred Model 4 under six alternative specifications and compare them with the 6NN baseline. The alternatives are four- and eight-nearest neighbors, queen and rook contiguity, and inverse distance and

[17] The regularity is about 2% per year across 1,528 regions in 83 countries (Gennaioli et al. 2014), and a similar figure across regions of the United States, Japan, and five European countries (Sala-i Martin 1996).

inverse-distance-squared, the last two applied within a distance band.

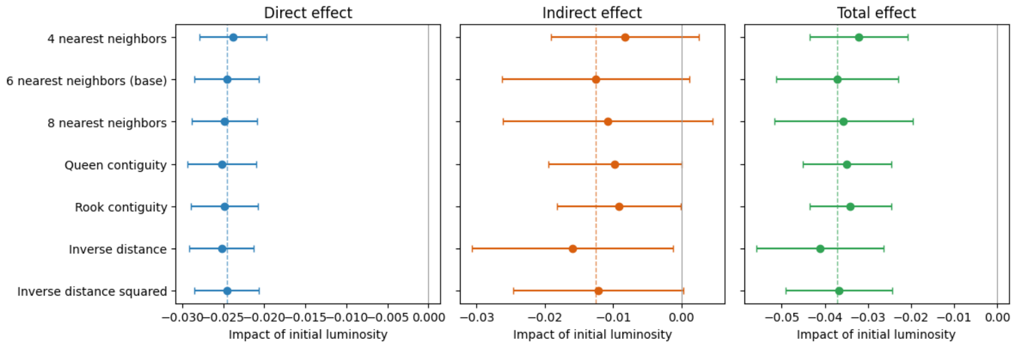


Figure 8: Robustness of the Model 4 spatial impacts of initial luminosity to the choice of spatial weight matrix. Points are Direct, Indirect, and Total impacts; bars are 95% Monte-Carlo confidence intervals. The dashed lines mark the 6NN baseline estimates.

Table 4: Model 4 spatial impacts of initial luminosity under alternative spatial weight matrices (full LeSage–Pace method; Monte-Carlo standard errors in parentheses).

Weight matrix	Direct	Indirect	Total	AIC
4 nearest neighbors	-0.024***(0.002)	-0.008(0.006)	-0.032***(0.006)	-2468
6 nearest neighbors (base)	-0.025***(0.002)	-0.013*(0.007)	-0.037***(0.007)	-2501
8 nearest neighbors	-0.025***(0.002)	-0.011(0.008)	-0.036***(0.008)	-2485
Queen contiguity	-0.025***(0.002)	-0.010**(0.005)	-0.035***(0.005)	-2463
Rook contiguity	-0.025***(0.002)	-0.009**(0.005)	-0.034***(0.005)	-2469
Inverse distance	-0.025***(0.002)	-0.016**(0.007)	-0.041***(0.008)	-2486
Inverse distance squared	-0.025***(0.002)	-0.012*(0.006)	-0.037***(0.006)	-2485

The convergence spillovers are robust to the choice of spatial weights (Figure 8 and Table 4). The direct (within-district) convergence effect is essentially unchanged across all seven specifications, ranging from -0.024 to -0.025 and always significant at the 1% level. The total convergence effect likewise remains negative and significant throughout, ranging from -0.032 to -0.041, with the 6NN baseline (-0.037) near the middle of this range. The indirect (spillover) component is negative in every specification and statistically significant under most of them. This stability indicates that the direct and total convergence effects are not an artifact of the particular neighbor definition. The spillover component is negative throughout and significant under five of the seven, which is a consistent if weaker pattern.

The results reported so far treat luminosity as a proxy for economic output, which is the use that the convergence literature has made of it. The discussion that follows asks first what the magnitude of these estimates implies for regional policy, and what rests on the use of a single growth period. It then turns to a second use of the same data, since nighttime lights also record electrification, urbanization, and broader socioeconomic conditions, and the reproducible workflow assembled here can be pointed at those dimensions as readily as at output. That possibility is pursued in an exploratory way, asking how luminosity relates to cultural participation across Indian states.

4 Discussion

4.1 *What the magnitude of the effects implies for regional policy*

The difference between the two readings of Model 4 is large enough to matter for policy. Under the OLS estimate, a district closes half of the gap to its steady state in about 23 years; under the spatial Durbin estimate, in about 13 years. Over the fourteen years of our sample, that is the difference between closing roughly 34% of the initial gap and closing roughly 52% of it. The first figure describes catch-up over a generation, while the second describes catch-up within the horizon over which development programs are designed, funded, and evaluated.

The distinction matters because it changes what can reasonably be expected of an intervention. If convergence is slow, support for lagging districts must be sustained across political cycles before its effects become visible in aggregate outcomes, and the absence of visible catch-up within the lifetime of a program is weak evidence that the program has failed. If convergence is faster, as the spatial estimates imply, the returns to a well-targeted intervention should appear within a normal evaluation window, and the case for measuring them is correspondingly stronger.

The results also speak to the level at which such policies should operate. We find convergence across districts over 1996–2010, whereas Lolayekar, Mukhopadhyay (2019), working with Indian states over a longer period, find divergence. Catch-up that is visible at the district level and absent at the state level is an argument for treating the district as the unit of intervention, which is what the Aspirational Districts Programme of India already does.[18] The spatial results add a qualification to that logic: the growth of a district is associated with the initial conditions of its neighbors, so a lagging district surrounded by other lagging districts is in a different position from one adjacent to a growing cluster, and a targeting rule that ranks districts one at a time will not register that difference.

Spillovers, however, cut both ways, and the Indian evidence on place-based policy is a caution as much as an encouragement. Hasan et al. (2021) evaluate a tax-exemption program for industrially backward districts and find that the resulting increase in firm entry and employment was driven in part by the displacement of activity from neighboring districts that narrowly missed qualifying, and that the effects did not persist once the program ended. Activity moved across a boundary would appear in our framework as a spatial relationship between neighbors, and it is not the kind of spillover on which a case for coordinated regional policy can be built. What the satellite record does support is measurement: luminosity is responsive enough to register a place-based package at this scale.[19]

These implications follow if the indirect effects reflect genuine interaction between districts. As set out in the results, the reduced-form evidence is consistent with that reading but does not establish it. The policy conclusions above should therefore be read with that qualification attached.

4.2 *The single growth period and what a panel would require*

The estimates reported above rest on a single growth period, and the cost of that design should be stated plainly. Three things follow from working with one long difference. It cannot trace how the rate of convergence evolved within the period. It cannot establish whether the spatial spillovers estimated above were stronger in some years than in others, or whether they emerged only part-way through. And it cannot separate long-run convergence from whatever was specific to these fourteen years, which in India included the maturing of the post-1991 reforms.

These are real limitations and the dynamics they concern are important, but they are beyond the scope of this article, whose purpose is to add a spatial dimension to

[18] The programme was launched in 2018 to direct additional resources to the least developed districts of the country.

[19] Shenoy (2018) uses nighttime lights to evaluate a large place-based package in Uttarakhand and finds that luminosity rises sharply on the targeted side of the new state border, which shows that the outcome we study is responsive enough to register policy at this scale.

an existing cross-sectional result rather than to replace it with a dynamic one. The immediate obstacle is the data. The dataset we adopt from Chanda, Kabiraj (2020) holds radiance-calibrated luminosity at six dates between 1996 and 2010, but district population is observed only at the decennial censuses. Per-capita luminosity at the intermediate dates would therefore rest on interpolated denominators. Over a single long difference that interpolation affects only the two endpoints, whereas in a panel it would enter every observation, and it would do so in the time dimension that a panel exists to exploit.

The more interesting question is what such a panel would be worth, and here two distinct problems compound. The first belongs to growth econometrics and predates nighttime lights. Panel estimates of convergence run far above cross-sectional ones: Caselli et al. (1996) obtain a rate near 10% per year using panel methods that instrument for endogenous regressors, against the 2% that cross-sections deliver. How to read that gap remains contested.[20] A panel specification would therefore not simply produce a better-identified version of our estimate; it would produce an estimate of something different, whose relation to the long-run rate is itself disputed.

The second problem is specific to the proxy, and it points the same way. Luminosity discriminates differences across places considerably better than changes over time (Chen, Nordhaus 2019, Asher et al. 2021). Convergence, however, is a long-run process, so the horizon over which it is measured determines whether lights carry the relevant signal at all. Over fourteen years most of the identifying variation is cross-sectional, which is the use that lights currently support best. Over the annual intervals that make panel methods attractive, the ratio of signal to measurement error in the light series falls sharply, so a short-horizon panel would combine a specification whose estimates are already hard to interpret with a proxy operating where it is weakest. Chakrabarty, Mukherjee (2022) estimate district convergence across the pandemic period, and how such estimates should be read depends on how much of the short-run movement in lights reflects output rather than measurement.

Our own view follows from this. If panel methods are to be applied to convergence in luminosity, the periods compared should be long: a sequence of non-overlapping long differences rather than a high-frequency panel. Harmonized DMSP–VIIRS series (Li et al. 2020) now make that feasible across roughly three decades, and re-estimating this specification as such a sequence is the most promising temporal extension of the analysis reported here.

4.3 An exploratory extension: Luminosity and cultural participation

Nighttime lights capture not just GDP but a composite of electrification, urbanization, infrastructure, and broader socioeconomic conditions (Henderson et al. 2012, Mellander et al. 2015). This multi-dimensional nature invites exploration of how luminosity relates to socioeconomic indicators beyond strictly economic output. In this section, we examine the association between luminosity and cultural participation patterns across Indian states.

A growing literature argues that regional cultural attitudes constitute an independent factor in economic development, not merely a by-product of income or institutions. In particular, Tubadji (2025) applies the Culture-Based Development framework, whose central claim is a directional one. Cultural attitudes and the participation patterns that reveal them are treated as a factor shaping local development, rather than as an outcome that development produces. That direction is what distinguishes the framework from accounts in which culture is a by-product of income, and it also marks the limit of what the exercise below can establish.[21]

For India, the National Sample Survey (NSS) 47th Round (July–December 1991) surveys household cultural participation. Following the Culture-Based Development

[20] Barro (2015) argues that country fixed effects generate a misleadingly high convergence rate, and that combining post-1960 with post-1870 evidence returns a figure close to the conventional 2%. Meta-analytic evidence finds that corrections for unobserved heterogeneity raise estimated convergence systematically (Abreu et al. 2005).

[21] A cross-sectional association is consistent with either direction, and with both operating at once. What it can do is establish that the two are related, and describe the form that relationship takes.

framework, we reduce these survey items to six dimensions and aggregate them to state means.[22] With nighttime lights data for an adjacent period (1992) now available for 32 Indian states and union territories, we can examine the extent to which luminosity is associated with regional cultural patterns. This part of the analysis uses the Consistent and Corrected Nighttime Lights series rather than the radiance-calibrated composites used above, because it is the product that covers 1992. Given the small sample size (N = 32) and the potential leverage of small territories at the extremes of the distribution on linear correlation measures, we report Spearman rank correlations throughout this section. Figure 9 presents the relationship between log nighttime lights per capita and the two cultural dimensions that show statistically significant associations.

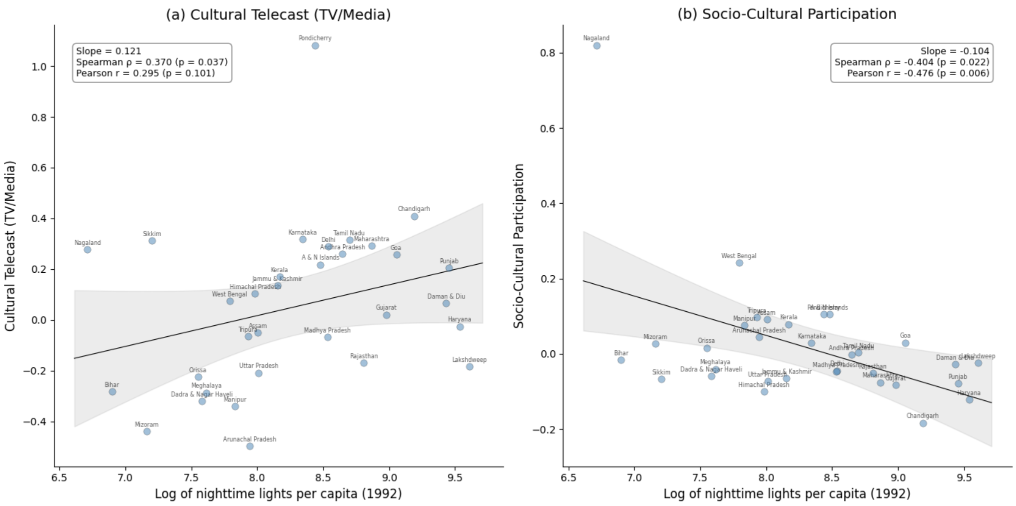


Figure 9: Relationship between nightlight luminosity and cultural participation across 32 Indian states Notes: Each point represents one of the 32 Indian states and union territories. Panel (a) shows the bivariate relationship between log nighttime lights per capita and cultural telecast (TV/media); Panel (b) shows the relationship with socio-cultural participation. Solid line is the OLS regression; gray band shows the 95% confidence interval (t-distribution, 30 df). Annotations report regression slope, Spearman rank correlation, and Pearson correlation. Source: Nighttime lights from CCNL DMSP-OLS (Zhao et al., 2022). Cultural participation from NSS 47th Round (July–December 1991). See Spatial culture notebook for source code.

Cultural telecast (TV/media) exhibits a positive association with nighttime light luminosity, with a Spearman rank correlation of 0.370 ($p = 0.037$). States with higher luminosity therefore consume more culture through television and media. This link is intuitive, because access to television depends on electrification and urbanization, which are closely tied to luminosity.

In contrast, socio-cultural participation shows a significant negative association, with a Spearman rank correlation of $-0.404(p$ = 0.022). Community-based cultural engagement is therefore stronger in states with less luminosity. This suggests that less urbanized regions rely more on collective, in-person forms of cultural participation than on media infrastructure.

Figure 10 presents the spatial distribution of these two cultural variables through the lens of a LISA analysis. Both dimensions exhibit significant spatial autocorrelation, which indicates that cultural participation is not randomly distributed across Indian states. The low-low cluster of cultural telecast in the northeast coincides with the low-low cluster in the nighttime lights data, although the high-high clusters of the two variables do not coincide. The significant high-high cluster in cultural telecast is confined to the southern

[22]The six dimensions are live cultural performance, cultural telecast (TV/media), socio-cultural participation, cultural heritage and religion, live cultural shows, and sports. They are extracted from the survey items by principal-components analysis. See the Spatial culture notebook for details and extended analyses.

states, while the northeastern states form a distinct high-high cluster in community-based participation.

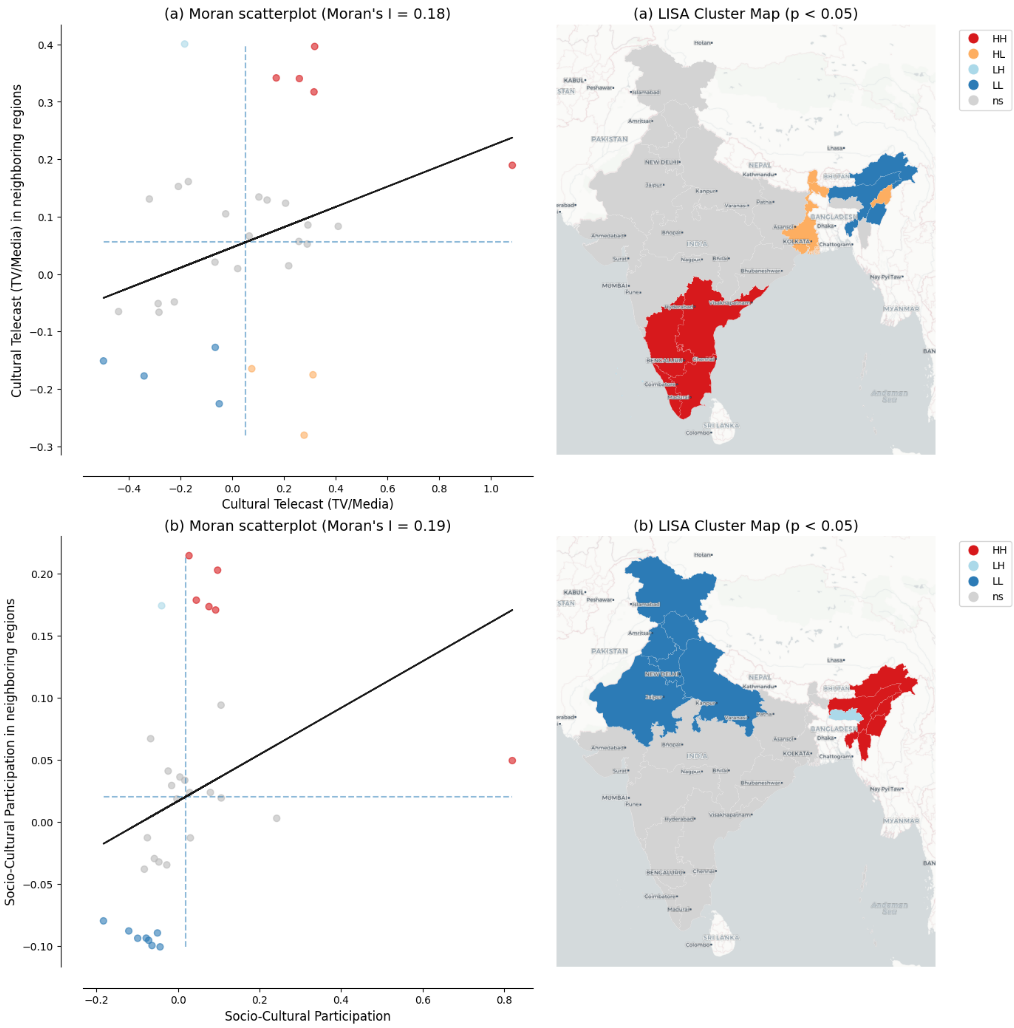


Figure 10: LISA cluster maps of cultural participation across 32 Indian states Notes: Panel (a) shows results for cultural telecast (TV/media); Panel (b) for socio-cultural participation. Left subpanels show Moran scatterplots with the Global Moran's I statistic. Right subpanels show LISA cluster maps with statistically significant clusters at $p < 0.05$ based on 999 permutations and a 6-nearest-neighbors spatial weights matrix. Region labels are overlaid from the CartoDB Positron basemap. Source: Cultural participation data from NSS 47th Round (July–December 1991). See Spatial culture notebook for source code.

Together, these two results reveal a contrast in how regions engage with culture: more luminous states favor media-based consumption, while less luminous states sustain community-based participation. The remaining four cultural dimensions show no statistically significant association with luminosity. These findings rest on a small cross-section of 32 states observed at a single point in time, so larger datasets covering more regions and periods would be needed to establish whether the patterns are robust.

Two connections to the main analysis are worth drawing out, both offered as interpretation rather than as findings. The first concerns what luminosity measures. A positive association with media-based cultural consumption and a negative one with community-based participation are what one would expect if luminosity tracks electrification and urbanization rather than economic output alone. The cultural results are, in that sense, a small piece of evidence about the composite nature of the proxy on which the convergence analysis rests, and they bear on the measurement issues discussed below.

The second connection concerns spatial structure. Cultural participation is not distributed randomly across states, since both dimensions examined here are significantly spatially autocorrelated. Factors of this kind, spatially clustered and correlated with development, are precisely what a spatial Durbin model absorbs into its indirect effects when they are not measured directly. We cannot test this with state-level data for a single adjacent period, and we do not claim that cultural participation drives the spillovers estimated in the results section. What the exercise does is make concrete what a reduced-form spatial model may be capturing, and it suggests that the Culture-Based Development framework offers one route to testing the question directly, should comparable data become available at the district level and over time.

4.4 New research directions

Three kinds of gap limit what this literature can currently establish, and each defines an agenda that the workflow used here is well placed to support. The first two concern the raw material, namely the data themselves and the geography those data cover. The third concerns the range of theories those data have been used to test.

4.4.1 Data gaps

Our analysis, following Chanda, Kabiraj (2020), relies on radiance-calibrated DMSP-OLS nighttime lights for 1996 to 2010, a dataset widely used as a proxy for economic activity (Henderson et al. 2012, Chen, Nordhaus 2011). DMSP-OLS is subject to well-documented limitations: top-coding in bright urban cores, the absence of on-board calibration, and blooming artifacts that spread light beyond its true source (Gibson et al. 2021, Abrahams et al. 2018). These are not merely nuisances, and it is worth being explicit about how they bear on our own estimates. Top-coding compresses variation at the bright end of the distribution, which affects the recorded initial luminosity of the most luminous districts. Two opposing biases follow: classical measurement error in the regressor attenuates the convergence coefficient toward zero, while measurement error in initial luminosity that enters the growth rate with the opposite sign works in the direction of overstating convergence. The net effect is ambiguous in sign, and we do not claim to sign it. Measurement error is also unlikely to be homogeneous across the sample, because the association between luminosity and economic outcomes is stronger in urban than in rural areas and Indian districts differ considerably in how urbanized they are. Blooming has a clearer implication for the spillover estimates that are central to this article. Because light spills across district boundaries, part of what we measure as the luminosity of a district originates with its neighbors, which mechanically raises the apparent correlation between adjacent districts and could therefore inflate the estimated indirect effects. That the direct and total effects hold across all seven definitions of the neighborhood, and the indirect effect under most of them (Table 4), shows they are not an artifact of any particular neighbor definition, but it does not rule out blooming; deblurring methods such as those of Abrahams et al. (2018) offer one route to addressing it directly.

The Visible Infrared Imaging Radiometer Suite (VIIRS), operational since 2012, improves on DMSP-OLS in spatial resolution (roughly 750 meters against 2.7 kilometers), in on-board radiometric calibration, and in dynamic range, avoiding saturation in bright areas while detecting dim lights in rural settlements (Elvidge et al. 2017, Gibson et al. 2021). The discontinuity between the DMSP (1992–2013) and VIIRS (2012–present) sensor eras remains an obstacle to long-run work, which harmonization efforts have begun to address (Li et al. 2020), and recent global series adjust explicitly for top-coding, blooming, and changes of satellite system (Chiovelli et al. 2026). Beyond better sensors, combining lights with other satellite products improves measurement where lights alone are weak.[23] Applying such multi-source approaches to Indian districts, and re-estimating

[23] Lights have been combined with daytime imagery for poverty prediction (Jean et al. 2016), with land cover in agricultural areas (Keola et al. 2015), with multiple remote sensing indicators for subnational output (Chen et al. 2024, Suleiman et al. 2025), and with survey data for multidimensional poverty (Khoun et al. 2025).

our specification on harmonized or VIIRS-era data, are the most immediate extensions of the analysis reported here.

4.4.2 Geographic gaps

A distinctive advantage of nighttime lights is that a single instrument observes the entire planet under a common protocol, so that measurements are comparable across countries and across regions in a way that national accounts, compiled under different statistical systems and capacities, are not (Henderson et al. 2012). Comparability across settings and scales nevertheless requires explicit adjustment (Chiovelli et al. 2026). This global coverage is, however, coverage of light rather than of economic activity, and the distinction determines where the method works. Roughly one fifth of the settlement footprint of the planet emits no radiance detectable from space, and unlit settlements are concentrated in Africa, which accounts for 39% of them, rising to 65% when only rural settlement areas are considered (McCallum et al. 2022). Lights therefore measure development that is illuminated: they track urban areas as they expand, and rural areas as lighting infrastructure reaches them, but they are close to silent where neither is happening. The empirical record matches this description.[24]

The resulting gap is not simply that some regions have gone unstudied. Regional inequality and convergence have been examined with luminosity across 49 African countries (Kacou 2022), as they have across Chinese provinces (Glawe, Mendez 2024), Indonesian districts (Miranti, Mendez 2023), Thailand (Tipayalai, Mendez 2024), and Turkey (Ursavaş, Mendez 2023). The gap is that the places where lights carry the least information, rural and sparsely electrified areas, are the same places where administrative statistics are weakest, which is precisely where a satellite proxy would be most valuable. Closing it calls less for further applications of the same proxy than for validation against ground data in those settings, and for the complementary sensors and multi-source methods described above.

4.4.3 Theoretical gaps

Within the convergence literature itself, our cross-sectional framework documents an average effect and its spillover component but cannot capture heterogeneity across the distribution. One direction is distributional: methods designed for that purpose could reveal whether Indian districts converge to a single steady state or form distinct clubs, and combining them with spatial econometrics could expose spatial poverty traps or growth poles that average estimates render invisible (Rey, Montouri 1999).[25] A second direction concerns identification: our spatial Durbin estimates capture spillovers in reduced form without establishing whether they operate through infrastructure linkages, labor migration, technology diffusion, or market access, and quasi-experimental designs such as that of Asher, Novosad (2020) offer a route to isolating them.

Beyond convergence, luminosity is now used to study city size and urban growth (Ch et al. 2021), electrification and the provision of public goods (Min et al. 2013), and armed conflict (Witmer, O'Loughlin 2011). As in the exploratory extension reported above, the same data can also be turned on socioeconomic dimensions that are not economic output at all, such as the cultural participation theorized by Tubadji (2025). A rapidly growing use is impact evaluation, in which lights measure the local effect of a policy or a shock at fine spatial scales (Bluhm, McCord 2022). Here a caveat is in order, and it is one the literature has established rather than one we are proposing. As noted above, lights discriminate differences across places considerably better than changes over time.[26] The responsiveness of lights to output also varies with baseline income,

[24] Across 34 sub-Saharan African countries the association between harmonized luminosity and household wealth is pronounced in urban areas and considerably weaker in rural ones (Jhamb et al. 2025), and systematic assessments of which product to use, and where, reach similar conclusions (Gibson et al. 2021).

[25] The relevant tools are distribution dynamics (Quah 1996) and the regression-tree methods developed for identifying multiple growth regimes (Durlauf, Johnson 1995).

[26] VIIRS predicts cross-sectional GDP better than time-series GDP (Chen, Nordhaus 2019). In India, the elasticity of lights with respect to local outcomes is far lower in the time series than in the cross-section (Asher et al. 2021). The relationship between lights and growth is also unstable across regions in both

population density, and sector (Bluhm, McCord 2022).[27] Evaluations over short horizons that rest on luminosity therefore require validation against local ground data before their estimates can be interpreted. The design used here, a fourteen-year horizon, rests mostly on cross-sectional variation, which is what lights currently support best.

Taken together, these three gaps describe a research agenda rather than a list of caveats, and none of the work they imply requires starting from scratch. The pipeline behind this article, running from the raw satellite composites through the construction of the spatial weight matrices to the impact decomposition and its robustness checks, is public, documented, and executable in a browser. A reader who wants to ask whether the patterns documented here hold in another country, in a later period, under a different definition of the neighborhood, or for an outcome other than economic output, can open the relevant notebook, substitute their own data, and re-run it. The appearance of spatial dependence in convergence processes across very different institutional settings also suggests that the questions raised here are not specific to India. That is the contribution we would most like this article to make: not only an estimate of how quickly Indian districts converged between 1996 and 2010, but a working template that lowers the cost of asking the same question somewhere else.

4.5 Research reproducibility and open science

The complexity of satellite-based economic research, which involves multi-step processing pipelines, spatial econometric estimation, and geographic visualization, makes reproducibility both challenging and essential. As Donaldson, Storeygard (2016) note, methodological choices in processing remotely sensed data, such as sensor inter-calibration, can affect the empirical conclusions. The same applies further down our own pipeline, at the construction of the spatial weight matrix. Our own robustness analysis is a case in point: the estimated convergence effects survive seven definitions of the neighborhood, but that is something a reader can only confirm because the code that produced each of them is available.

Open-source tools and cloud platforms are lowering the barriers to this kind of verification. Cloud-based notebooks, such as those developed by Patnaik, Mendez (2024) for processing nighttime lights data, remove the need for specialized local software. The combination of open data, version-controlled code, and cloud execution makes the full pipeline, from satellite imagery to econometric results, verifiable and extensible by the broader research community.

5 Concluding remarks

This article re-examines the regional convergence hypothesis across Indian districts using satellite nighttime light data, interactive visualizations, and spatial econometric modeling. Building on the work of Chanda, Kabiraj (2020), we developed an interactive web-based visualization tool, and our spatial autocorrelation analyses indicate that spatial dependence is a notable characteristic of both the satellite data and the regional convergence process in India. Estimates from our spatial Durbin model indicate that incorporating spatial spillovers increases the estimated speed of regional convergence. The total convergence effect in our fully specified model is approximately 51% larger than conventional non-spatial estimates, and the implied half-life of regional disparities falls from roughly 23 years to 13. This finding suggests that in this setting non-spatial convergence models understate the speed of regional convergence, although the comparison with other regional literatures shows that the direction of this adjustment is specific to the setting rather than general. If the estimated indirect effects reflect genuine spillovers, place-based development interventions would have broader impacts than conventional estimates suggest, since their benefits would reach neighboring districts as well as the districts they target. If instead they partly reflect activity displaced across district boundaries, as the evaluation

Brazil and India (Bickenbach et al. 2016).

[27] An evaluation may therefore fail to detect a real effect where lights react weakly, or attribute to a policy what is in fact heterogeneity in the relationship between light and output.

literature on Indian place-based programs cautions, the same estimates would overstate what those interventions add in aggregate.

Our results also demonstrate the usefulness of satellite nighttime lights for studying economic dynamics in countries where subnational data are scarce, infrequent, or unreliable. Conventional economic statistics at the district level are often unavailable or inconsistent across administrative boundaries, especially in large developing economies. Radiance-calibrated nighttime light data allowed us to analyze 520 Indian districts at a spatial granularity that would be difficult to achieve with traditional national accounts. This data-driven approach is potentially transferable to other data-poor contexts across the developing world, and the ongoing transition from DMSP-OLS to the higher-resolution VIIRS sensor should yield more precise measurement of subnational economic activity in future studies.

This study also shows how reproducible open-science practices can strengthen the credibility and reach of scientific research. The entire analytical pipeline is documented in Jupyter notebooks written in Python, and the results are embedded within the manuscript through the Quarto publishing framework.[28] By hosting the complete codebase and data in a public GitHub repository, and by making the Python notebooks executable in the cloud through Google Colaboratory, we aim to encourage broader adoption of reproducible open-science practices. Section 4 sets out where we believe the most valuable work now lies, in better data, in the regions this method currently serves least well, and in the questions beyond convergence that it is able to address. The materials accompanying this article are meant to make starting on any of them straightforward.

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7 Conflict of Interest

The authors declare no conflict of interest.

8 Data and Code Availability

All data and computational code used in this study are available in the project repository at https://github.com/quarcs-lab/project2025s-py. The interactive HTML version of this manuscript (https://quarcs-lab.github.io/project2025s-py/) embeds the computational notebooks, allowing readers to inspect the complete analytical pipeline from raw data to published results. Every analytical notebook can be executed in the cloud using Google Colaboratory without any local software installation: readers open the notebook and run it, and the results, figures, and tables of this article are regenerated from the raw data. The computational notebooks are implemented entirely in Python. An earlier implementation of the same analysis used R and Stata, and the present Python results are largely consistent with it. The impact decomposition of the unconditional spatial Durbin model (Model 1) differs, although its total effect is unchanged, and the AIC values differ by a few units. The interactive Earth Engine application linked in the text is a convenience layer for visualizing the underlying imagery; its source code is included in the first notebook, and none of the results reported in this article depend on it.

[28] A single manuscript source generates multiple output formats. The HTML version allows readers to engage with interactive visualizations, inspect the code of the computational notebooks, and re-evaluate the results in light of their source code.

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